Predicting Falcon 9 First-Stage Landing Success

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Executive Summary

This project aimed to predict the success of Falcon 9 first stage landings, a critical contributor to SpaceX's cost efficiency through rocket reusability.

A number of data sources were available containing historical information on the landing success rate of the Falcon 9 first stage. Data collected from the SpaceX API and web-based sources such as Wikipedia was cleaned, transformed, and converted into DataFrames for further analysis and for building a machine learning model to predict the expected outcome.

Using the DataFrames generated with Python's pandas library, several key insights were identified during the Exploratory Data Analysis (EDA) phase of the project. Relationships were observed between multiple variables, such as the booster version used, payload mass, orbit flown, and the corresponding landing and launch sites.

Given the data analysis results and the defined business requirements, a classification machine-learning approach was deemed suitable for the prediction task. Models including Logistic Regression, SVM, KNN, Random Forest, and XGBoost were implemented and evaluated.

The model predictions predominantly indicated successful first-stage landings.

Introduction

SpaceX Falcon 9 Overview

The Falcon 9 is a medium-lift launch vehicle developed and manufactured by SpaceX, an American aerospace company founded by Elon Musk. Standing approximately 230 feet tall with a diameter of 12 feet, it is powered by nine Merlin engines on the first stage and a single Merlin engine on the second stage.

Falcon 9 operates on a two-stage system:

• First Stage (Booster): Provides initial thrust to lift the rocket off the ground. It can return to a landing pad or drone ship for reuse.

• Second Stage: Ignites after stage separation to deliver payloads to their target orbit.

Falcon 9 is notable as the first orbital-class reusable rocket. This reusability has dramatically reduced launch costs, making space access more affordable for commercial and government customers.

Project Statement: With the introduction of the Falcon 9 rocket, is it possible to accurately predict whether the first stage will land successfully in future missions?

Methodology

The standard Data Science Methodology was used in this project to address the objectives:

1. Business Understanding
2. Analytic Approach
3. Data Requirements
4. Data Collection
5. Data Understanding & Preparation
6. Exploratory Data Analysis (EDA)
7. Modeling
8. Model Evaluation

Business Understanding

The Falcon 9 booster project focuses on predicting the likelihood of successful first-stage landings.

- **Business Goal:** Improve mission planning and operational efficiency by anticipating landing outcomes.
- **Value Proposition:** Accurate predictions reduce risk, optimize resource allocation, and support engineering refinements.
- **Key Stakeholders:** SpaceX operations teams, engineers, and management who rely on reliable forecasts to guide launch decisions.
- **Success Criteria:** A predictive model that achieves high accuracy and recall, minimizing false negatives (missed successful landings) and false positives (incorrectly predicted successes).

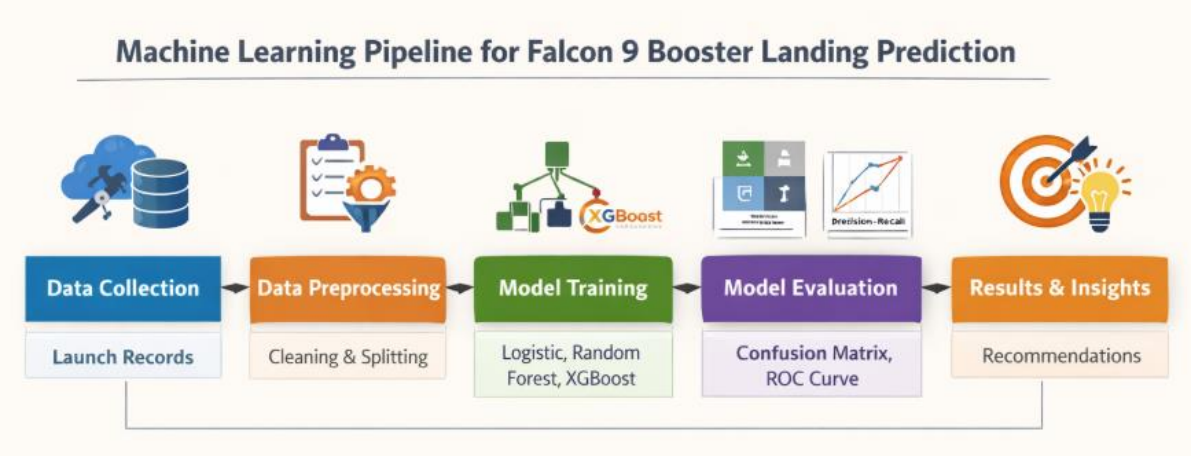
Analytic Approach

To address the business problem, a machine learning classification framework was applied:

1. **Data Preparation**
    - Historical launch records were cleaned, standardized, and split into training/testing sets.
    - Features included booster version, payload mass, orbit type, launch site, and weather conditions.
2. **Model Selection**
    - Logistic Regression for baseline interpretability.
    - Random Forest for robust ensemble predictions.
    - XGBoost for optimized performance on complex feature interactions.
3. **Evaluation Strategy**
    - Confusion Matrix to identify error types.
    - ROC Curve and AUC to measure classification strength.

- Precision, Recall, and F1-score to balance success/failure predictions.

4. **Insights Extraction**
   - Feature importance analysis highlighted booster version and payload mass as dominant predictors.
5. **Visualization & Communication**
   - Results presented via confusion matrix heatmaps, ROC curves, and stacked bar charts.



Results & Discussion

Models Tested:

• Logistic Regression – Interpretable, good baseline for binary classification.

• Support Vector Machines (SVM) – Useful if the dataset is balanced and not too large.

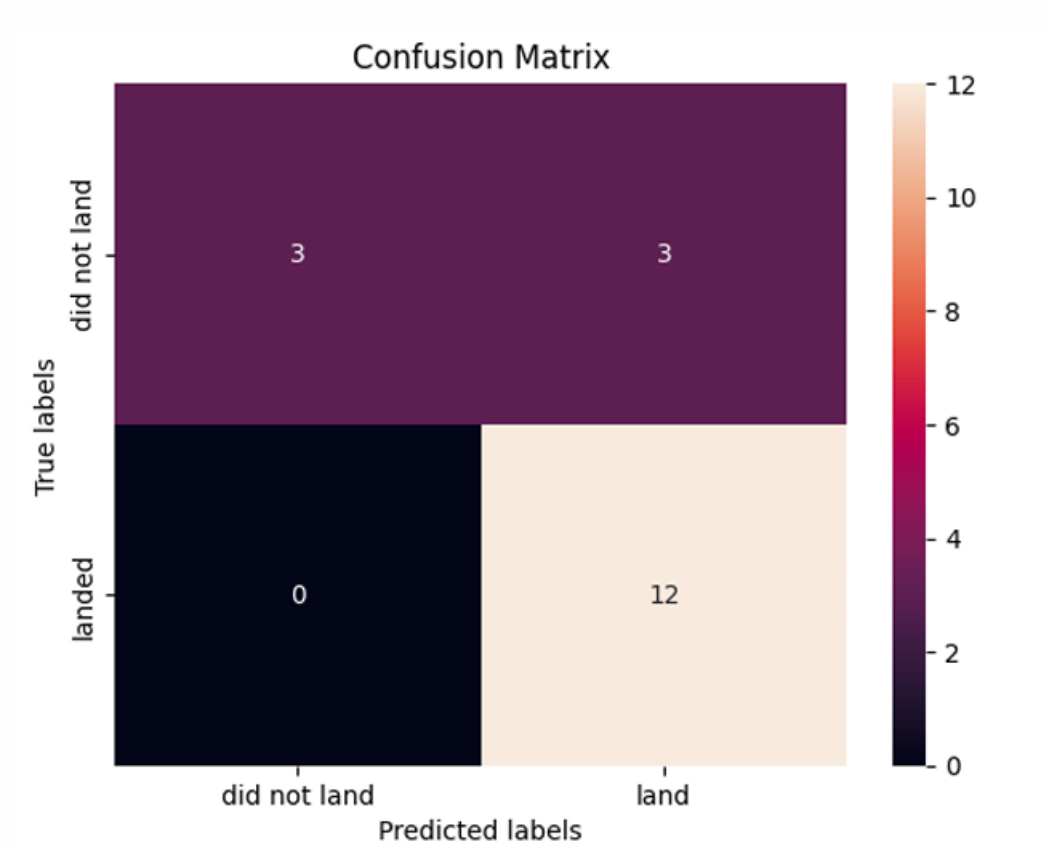
• Decision Tree – Handle categorical + continuous features well, capture nonlinear relationships.

• XGBoost – Powerful for tabular data, often yields high accuracy.

The table below shows a summary result of the accuracy performance of each classifier tested.

MODEL	CV SCORE	CLASSIFICATION REPORT SCORE	ROC SCORE
LOGISTIC REGRESSION	0.83	0.83	0.91
SVM	0.83	0.83	0.87
DECISION TREE	0.83	0.83	0.70
KNN	0.83	0.83	0.79
RANDOM FORREST	0.86	0.83	0.83
XGBOOST	0.86	0.83	0.89

Confusion Matrix Results – How well the classification model predicted the booster landing success vs failure.



For all the models tested the Confusion Matrix prediction results are the same.

Breakdown of Results:

• True Negatives (3) -> Correctly predicted “did not land”.

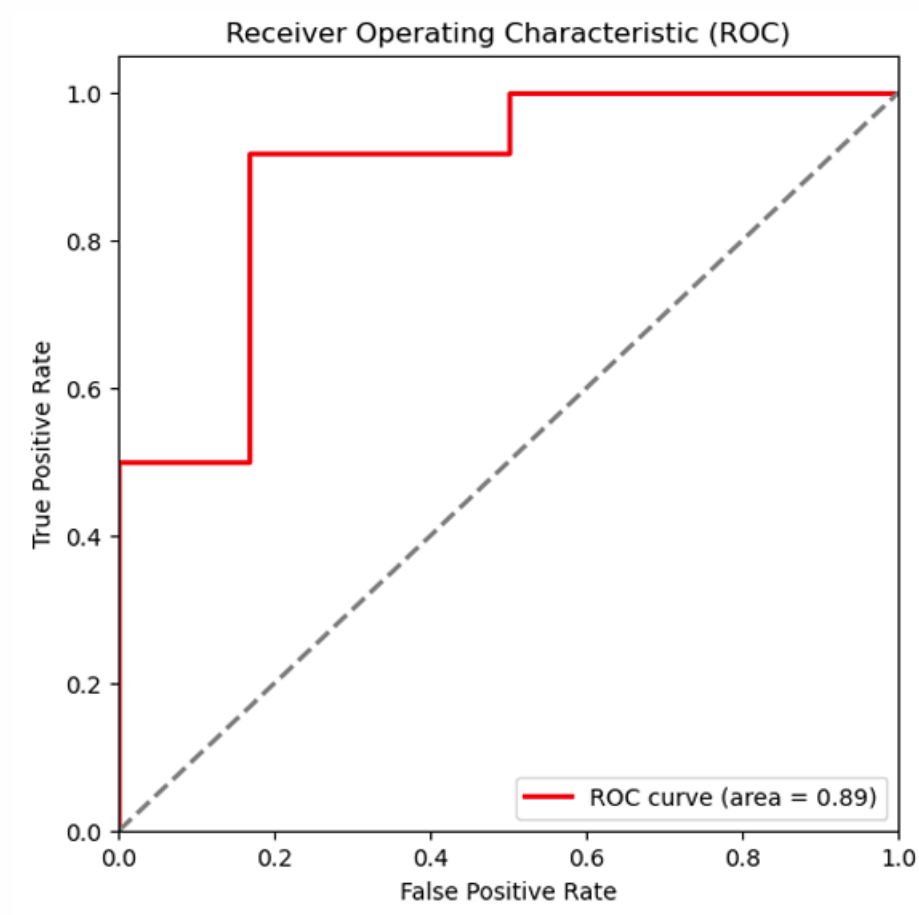
• False Positives (3) -> Predicted “landed” when the booster actually did not land.

• False Negatives (0) -> No cases where the model predicted “did not land” but the booster actually landed.

• True Positives (12) -> Correctly predicted “landed”.

For the Random Forrest and XGBoost classifiers a pipeline was used incorporating a Cross-Validation method and also added a ROC curve graph to gain more insight into the accuracy of these two models.

Below is the Receiver Operating Characteristics (ROC) curve for the selected XGBoost classifier.



The ROC curve with an AUC of 0.89 indicates that the classification model performs very well at distinguishing between successful and failed booster landings.

The added Cross-Validation method also proofed to be very effective and confirmed that ensemble methods provide more reliable predictions than single models.

Overall Outcome: Best balance: Logistic Regression (interpretability + high ROC) and XGBoost (strong CV + ROC). Robust ensemble: Random Forest

also performed well, though slightly weaker in ROC. Weaker options: Decision Tree and KNN showed lower discriminative ability.

Conclusion

Applying the data science methodology provided a structed and effective approach to addressing the problem of predicting SpaceX Falcon 9 first-stage landing success.

Beginning with a clear business and analytic understanding, the project identified rocket reusability as a critical factor influencing launch costs and competitiveness in the aerospace industry.

Data collection from the SpaceX API and supplementary Wikipedia sources ensured a comprehensive dataset, while systematic data wrangling improved data quality and enabled the creation of a meaningful target variable. It was event with the data collection process that only 91 rows of data was available which will impact predictive modelling outcome.

Exploratory data analysis revealed key relationships between landing success and factors such as payload mass, orbit type, and launch site, offering valuable domain insights. Interactive visualizations further enhanced understanding by highlighting geographic and operational patterns. From the data discovery, it is event that over time, the booster landing success rate improved, and one specific launch site was more prominent to the success of the SpaceX Falcon 9 program.

In the modeling phase, multiple classification algorithms were developed, tuned, and evaluated, demonstrating that machine learning techniques can reasonably predict booster first-stage landing outcomes using historical launch data. All models exhibited a high ability to discriminate between successful and unsuccessful landings, as indicated by strong ROC performance. Among the models tested, the XGBoost classifier consistently outperformed the others, making it the best-performing model for the dataset.

References

The following links contain information regarding the datasets and Jupyter notebooks used for this project:

- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/tree/main/Module%209

- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/tree/main/Module%2010